sterilization doses in the range 25–50 kGy have been adapted in many countries. Both ^{60}Co and high energy electron accelerators are used as radiation sources. Radiation sterilization has the advantage of avoiding the use of toxic chemicals (e.g. ethylene oxide) and high temperature. A further advantage is that the material can be sterilized in a continuous process after final packaging.

Wire and cable: Thermoplastics and elastomers are widely used as electrical insulating material due to their physical properties and processability. Cross linking is an effective means for improving e.g. the thermal resistance and tensile strength. EBA-irradiation ($\approx$50 kGy) affords a rapid, well controlled cross linking and is used by several major producers of thin wires and cables.

Shrinkable film and tubing: Cross linked semi-crystalline thermoplastics display rubberlike properties at temperatures above their melting points. On deformation followed by fast cooling the polymer maintains its deformed shape. The polymer returns to its original shape when reheated. This memory effect is applied in the production of heat shrinkable films and tubing. Radiation doses of the order of 40–100 kGy are used in the production of heat shrinkable products.

Curing of surface coatings and inks: Irradiation is used to cure printing inks and varnishes and to bond coatings to surfaces of paper, fabrics, polymer films and steel. The absorbed doses are in the range 10–100 kGy and use of accelerators with low energy and high beam current affords selective energy deposition in the surface layer and high product throughput. Industrial radiation curing lines are operating in the car, wood panel and steel industries. Due to the low electron energy, no heavy shielding is required and the accelerators can easily be placed in normal industrial environment.

Potential applications: Immobilization or trapping of bioactive material such as enzymes, antigens, antibodies and drugs in polymer matrixes by radiation processes have received much attention.

Radiation treatment: To remove organics and metallic pollutants from waste water and SO_x and NO_x from flue gases emitted from coal power stations and industrial plants have been studied in great detail. The cleaning of flue gases has been developed to the pilot plant stage.

8.12. TECHNICAL USES OF SMALL DOSE-RATES

The ability of nuclear radiation to ionize gases is used in several applications. Static electricity can be eliminated through the installation of an α- or β-source in microbalances. Similarly, but on a much larger scale, elimination of static electricity is used in the paper, textile and paint industries. An ionization instrument for the analysis of gas has been developed in which the gas passes through a small chamber where it is irradiated by a small radioactive source. For a constant source of radiation, the ions produced in the gas depend on the flow velocity of the gas and on its temperature, pressure and atomic

composition. The dependence of the ionization on the atomic composition is a consequence of the different ionization potentials of the different types of atoms of the gas and the different probabilities for electron capture and collision. The ion current is collected on an electrode and measured. This current is a function of the gas pressure and velocity since the higher the pressure, the more ions form, while at higher velocity, the fewer ions are collected as more ions are removed by the gas prior to collection. Such ionization instruments are used in gas chromatographs and other instruments as well as in smoke detection systems (the normal radiation source is ^{241}Am, usually $\leq$40 kBq), where secondary electrons condense on smoke particles, leading to lower mobility for the electrons and a decreased ion current.

Both α- and β-emitters are used in luminescent paint. The fluorescent material is usually ZnS. T and ^{14}C are preferred sources since their β-energies are low, but ^{85}Kr, ^{90}Sr, and ^{147}Pm are also used. The amount or radioactivity varies, depending on the need (watches, aircraft instruments, etc.) but it is usually <400 MBq (<10 mCi), although larger light panels may require >50 GBq (several Curies). For such high activities only T or ^{85}Kr are acceptable because of their relatively low radiotoxicity.

8.13. EXERCISES

8.1. How many ion pairs are produced in 10 m of air of STP by one (a) 5 MeV α-particle, (b) 1 MeV β-particle, and (c) 1 MeV γ-quantum (μ_m(air) ≈ μ_m(water))?

8.2. Estimate the fraction of energy lost through bremsstrahlung for a β-emission of $E_{max} = 2.3$ MeV, when absorbed in aluminum. The effective β-energy must be taken into account.

8.3. A freshly prepared small source of ^{24}Ne had a measured decay rate of 1 GBq 1 s after its preparation. The source is shielded by 10 cm of Pb. ^{24}Ne emits γ-rays; 8% with 0.878 MeV and 100% with 0.472 MeV. Estimate the total integrated dose received at 2 m distance during its life-time after preparation. Neglect build-up factors.

8.4. An acidic aqueous solution is irradiated by α-particles from dissolved ^{239}Pu at a concentration of 0.03 M. The plutonium is originally in its hexavalent state, but is reduced to the tetravalent state by the reaction Pu(VI) + 2H· → Pu(IV) + 2H$^+$. How much of the plutonium can be reduced in one week?

8.5. An acid solution of fresh fission products contains 0.8 g l^{-1} cerium as Ce^{4+}. The γ-flux in the solution corresponds to 520 GBq l^{-1} of an average energy of 0.7 MeV. If half of the γ-flux is absorbed in the solution, what fraction of cerium is reduced to Ce^{3+} in 24 h? Assume the same *G*-value as in Figure 8.7.

8.6. Estimate the average *LET* value in water for β-particles from ^{90}Sr and ^{90}Y, and in Al for T.

8.7. A ^{60}Co irradiation source is calibrated by the Fricke dosimeter for which the *G*-value is assumed to be 1.62 μmol/J. Before the irradiation the optical density *D* of